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Abstract

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1 Sec 1

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration - Macros are everywhere - Artificial intelligence

We delegate tasks to some tools.

In the domain of formal proof automation is everywhere, and it is formally called elaboration ??

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

1.1 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-language.

We also talk about a formalized tool. . .

2 Meta-programming

Tactics, coq-elpi

3 Type classes resolution in Elpi

Discrimination trees: opt for lists and tree depths.

The entry point. . .

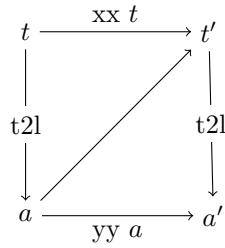
[1]

[2]

4 Determinacy checking for Elpi

hard cut as in prolog

[3]



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Discrimination trees: discriminate rules head

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
pred g o:int, o:int.
```

```
f 3 3 (g X X).
```

```
f 3 4 (g 3 4).
```

```
main:-
```

```
f X Y (g X Y).
```

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In the code above, both rules can be used with the query, but, the discrimination tree, save the world

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References

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